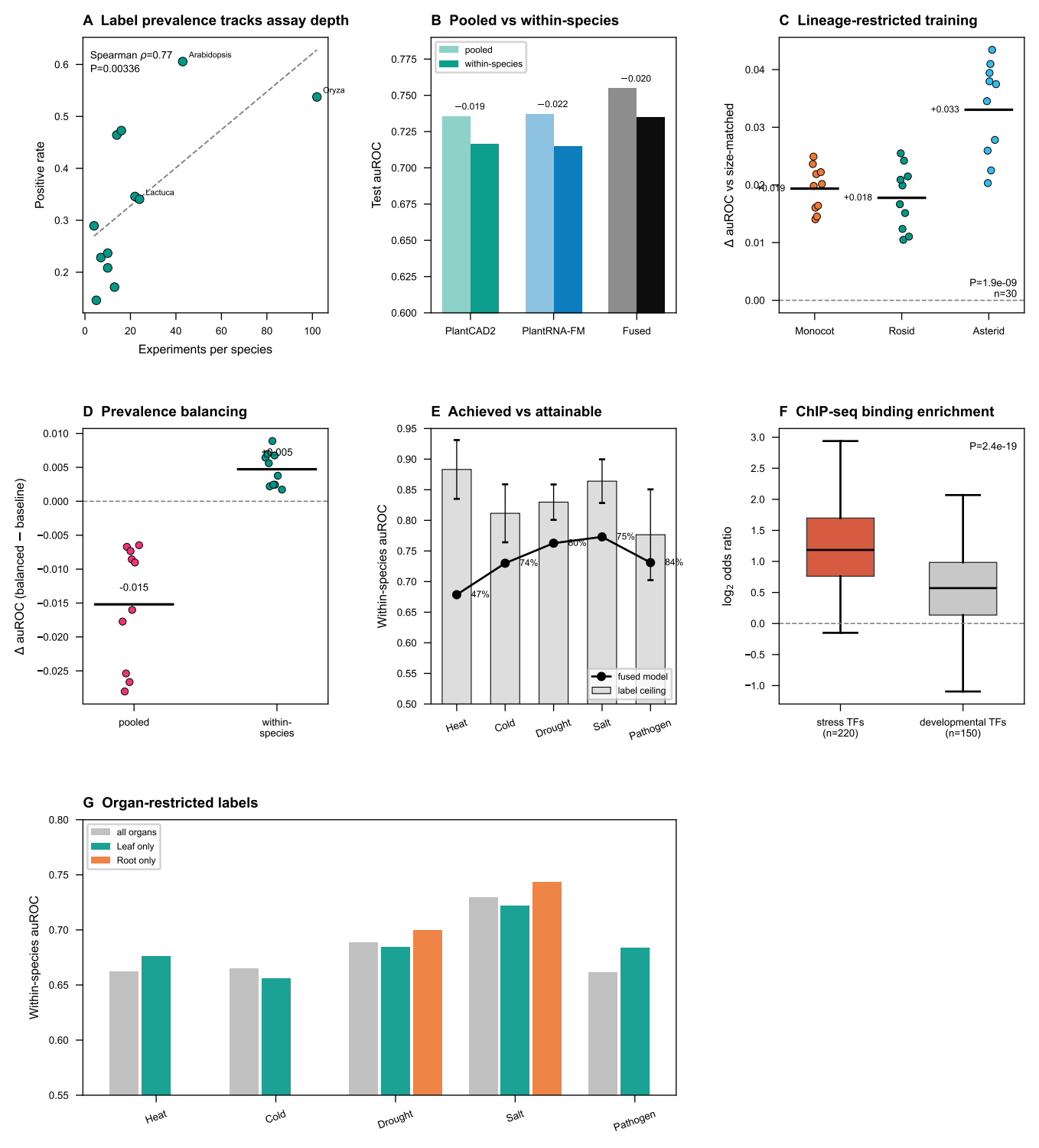
A Label prevalence tracks assay depth
Spearman ρ=0.77
P=0.00336
Arabidopsis
Oryza
Lactuca
Positive rate
Experiments per species
B Pooled vs within-species
pooled
within-species
Test auROC
−0.019
−0.022
−0.020
PlantCAD2
PlantRNA-FM
Fused
C Lineage-restricted training
Δ auROC vs size-matched
+0.019
+0.018
+0.033
P=1.9e-09
n=30
Monocot
Rosid
Asterid
D Prevalence balancing
Δ auROC (balanced − baseline)
+0.005
-0.015
pooled
within-species
E Achieved vs attainable
Within-species auROC
47%
74%
80%
75%
84%
fused model
label ceiling
Heat
Cold
Drought
Salt
Pathogen
F ChIP-seq binding enrichment
log2 odds ratio
P=2.4e-19
stress TFs
(n=220)
developmental TFs
(n=150)
G Organ-restricted labels
Within-species auROC
all organs
Leaf only
Root only
Heat
Cold
Drought
Salt
Pathogen